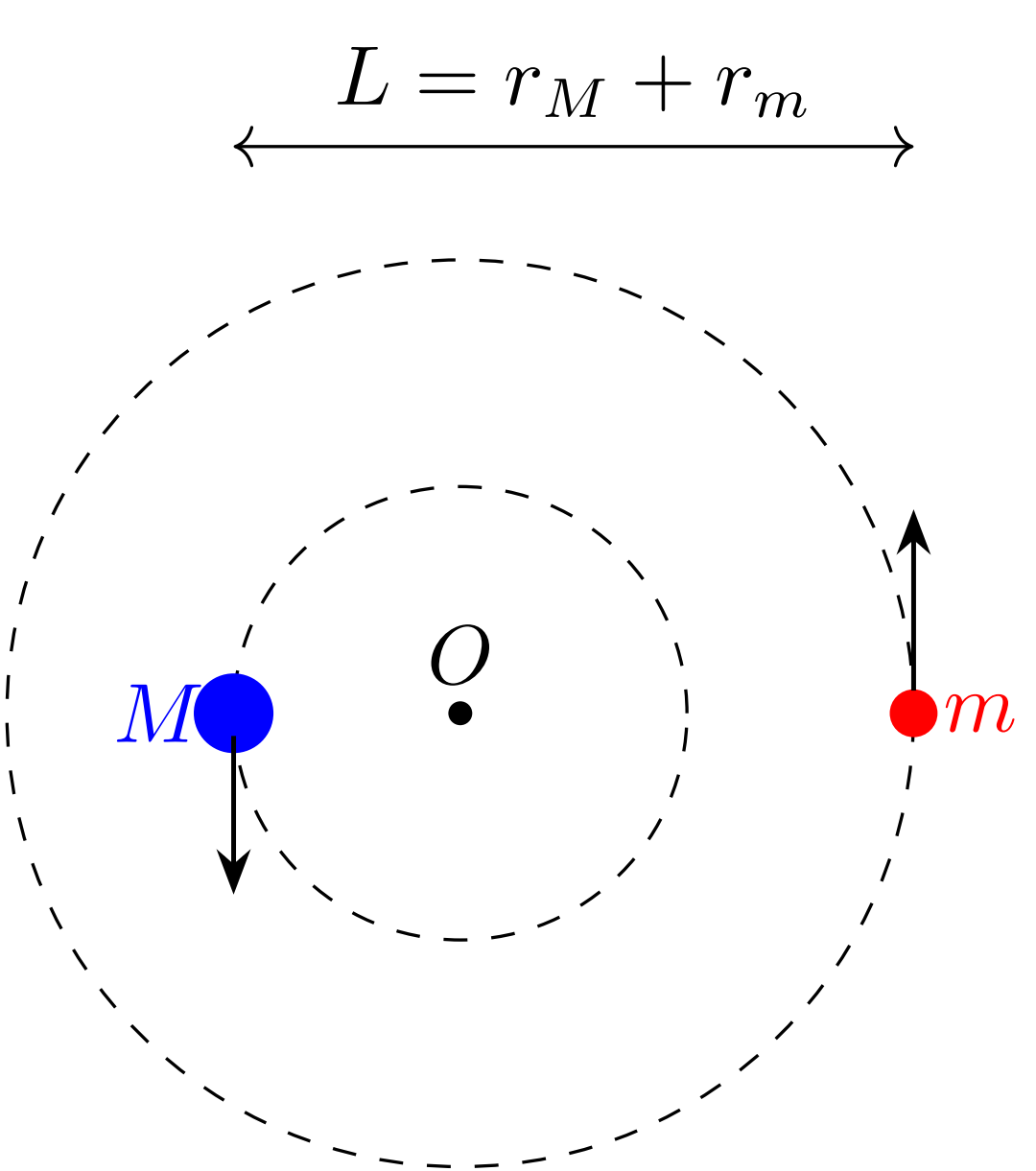

$$Mr_M = mr_m, \quad I = \mu L^2$$